BTS – IV – 04.14 – 0123

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B.Tech. Degree IV Semester Examination April 2014

ME 405 HYDRAULIC MACHINERY
(2006 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART A
(Answer *ALL* questions)

$(8 \times 5 = 40)$

I. (a) List and describe the three necessary conditions for complete similarity between a model and a prototype.

(b) Show that when a jet of water impinges on a series of curved vanes, maximum efficiency is obtained when the vane is semicircular in section and the velocity of the vane is half that of the jet.

(c) What are the functions of a draft tube?

(d) What is cavitation? How can it be avoided in a reaction turbine?

(e) What is meant by Net Positive Suction Head?

(f) What are the functions of an air vessel in a reciprocating pump?

(g) Write a short note on hydraulic coupling.

(h) Explain the working of a gear pump with a neat sketch.

PART B

$(4 \times 15 = 60)$

II. The drag force on a high speed aircraft depends on the velocity of flight V, the characteristic geometrical dimension of the aircraft l, the density ρ, viscosity μ and isentropic bulk modulus of elasticity E_s of ambient air. Using Buckingham's pi-theorem, find out the independent dimensionless quantities which describe the phenomenon of drag on the aircraft. (15)

OR

III. A series of curved vanes with entrance angle 30° and exit angle 15°, deflect a jet of water $10 cm^2$ in area, moving at 50m/s and inclined at 15° to the line of motion of the vane, Determine (15)

(i) the velocity of the vane to avoid shock at entry
(ii) the magnitude and direction of the resultant force on the vane and the force in the direction of motion
(iii) the magnitude and direction of velocity at exit.

(P.T.O.)

IV. A Pelton wheel nozzle, for which the coefficient of velocity $C_v = 0.97$, is 400m below the water surface of a lake. The jet diameter is 80mm, the pipe diameter is 0.6m, its length is 4km and $f = 0.008$. The buckets deflect the jet through 165° and they run at 0.48 times the jet speed, bucket friction reducing the relative velocity at outlet by 15 percent of the relative velocity at inlet. The mechanical efficiency of the turbine is 90%. Find the flow rate and shaft power developed by the turbine. (15)

OR

V. (a) What is specific speed? State its significance in the study of a hydraulic turbine. (4)

(b) Derive an expression for the specific speed of a turbine. (5)

(c) A turbine is to operate under a head of 30m at 300 rpm. The discharge is $10m^3/s$. If the efficiency is 90%, determine the specific speed of the turbine and the power developed. (6)

VI. (a) Derive an expression for the minimum speed for starting a centrifugal pump. (5)

(b) Draw the inlet and outlet velocity triangles of a typical centrifugal pump and derive an expression for the work done by the impeller on water per second per unit weight of water. (6)

(c) What are the different types of losses that occur inside the impeller of a centrifugal pump? (4)

OR

VII. A single acting reciprocating pump has a plunger diameter of 200mm and stroke of 300mm. The suction pipe is 100mm in diameter and 8m long. The water surface in the sump from which the pump draws water is 4m below the pump cylinder axis. If the pump is working at 30 rpm, find the pressure head on the piston at the beginning, middle and end of the suction stroke. Take friction factor, $f = 0.04$. (15)

VIII. Explain in detail the working of a hydraulic ram with the help of a neat diagram. (15)

OR

IX. Write short notes on: (15)

(i) Hydraulic press
(ii) Hydraulic accumulator
(iii) Hydraulic intensifier

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